UNIVERSITY OF INFORMATION TECHNOLOGY AND SCIENCES

Lab report

Course Title: Microprocessors and Microcontrollers Lab

Course Code: CSE-360

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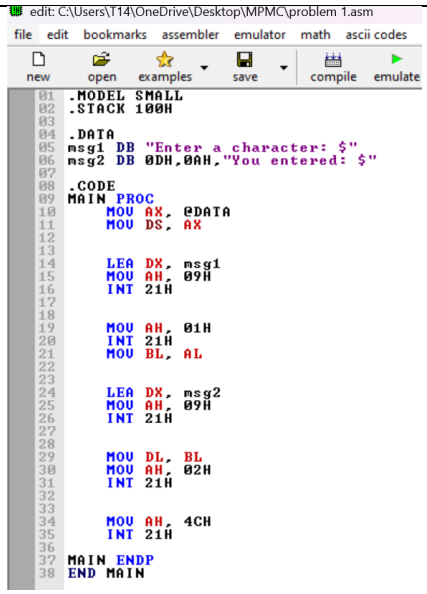
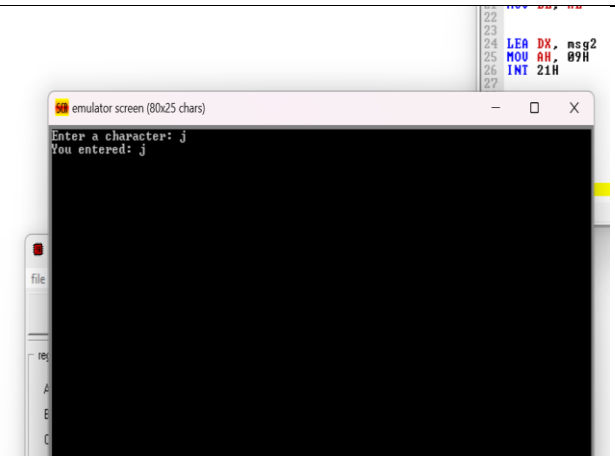
Experiment No: 01

Experiment Name: Write an Assembly Language program to take input from the user and display the output.

Introduction:

Assembly Language is a low-level programming language that helps us understand how a computer works internally. It uses simple instructions that are directly related to machine language. In this experiment, an Assembly Language program was written to take input from the user and display the same output on the screen. The program uses DOS interrupt functions for input and output operations. This experiment helps to understand basic Assembly concepts such as registers, interrupts, and data transfer between registers.

Code	Output
	

```
.MODEL SMALL
.STACK 100H

.DATA
msg1 DB "Enter a character: $"
msg2 DB 0DH,0AH,"You entered: $"

.CODE
MAIN PROC
    MOV AX, @DATA
    MOV DS, AX

    LEA DX, msg1
    MOV AH, 09H
    INT 21H

    MOV AH, 01H
    INT 21H
    MOV BL, AL

    LEA DX, msg2
    MOV AH, 09H
    INT 21H

    MOV DL, BL
    MOV AH, 02H
    INT 21H

    MOV AH, 4CH
    INT 21H

MAIN ENDP
END MAIN
```

```
Enter a character: j
You entered: j
```

Conclusion:

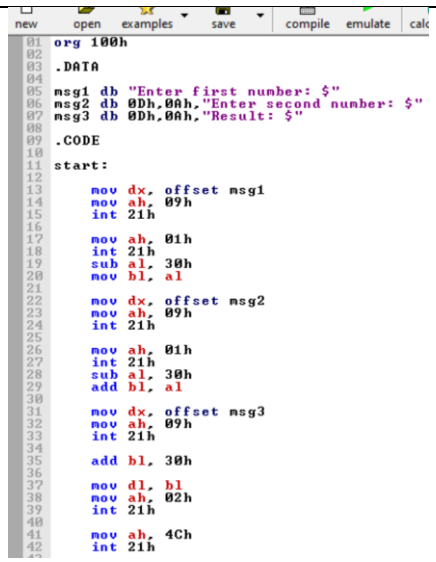
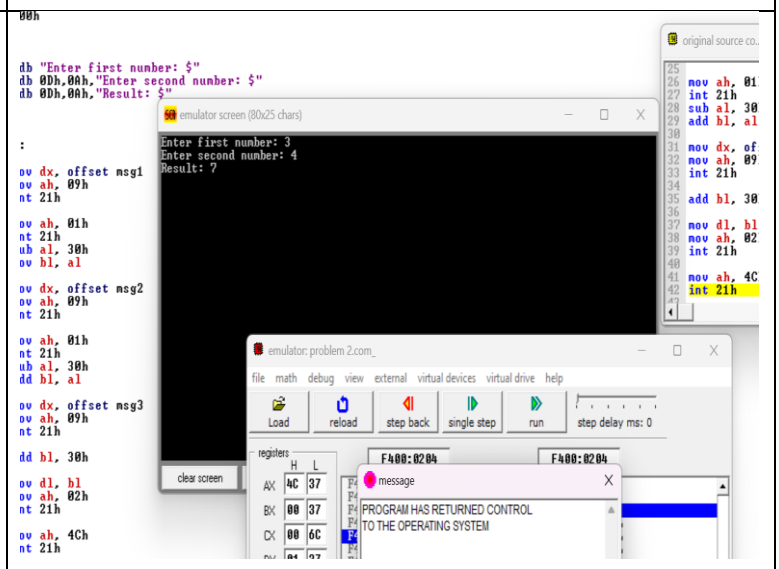
From this experiment, we learned how to take input from the user and display output using Assembly Language programming. We also learned the use of registers and DOS interrupt services for performing input and output operations. The program was successfully executed in EMU8086.

Experiment No: 02

Experiment Name: Write an Assembly Language program to perform addition of two numbers.

Introduction:

Assembly Language is a low-level programming language that is used to communicate directly with computer hardware. It helps programmers understand the internal working process of a computer system. In this experiment, an Assembly Language program was developed to perform the addition of two numbers. The program takes two numbers as input from the user, adds those using Assembly instructions, and displays the result on the screen. This experiment helps to understand basic arithmetic operations, register usage, and input-output functions in Assembly Language.

Code	Output
	

```
org 100h

.DATA

msg1 db "Enter first number: $"
msg2 db 0Dh,0Ah,"Enter second number: $"
msg3 db 0Dh,0Ah,"Result: $"

.CODE

start:

    mov dx, offset msg1
    mov ah, 09h
    int 21h

    mov ah, 01h
    int 21h
    sub al, 30h
    mov bl, al

    mov dx, offset msg2
    mov ah, 09h
    int 21h

    mov ah, 01h
    int 21h
    sub al, 30h
    add bl, al

    mov dx, offset msg3
    mov ah, 09h
    int 21h

    add bl, 30h

    mov dl, bl
    mov ah, 02h
    int 21h

    mov ah, 4Ch
    int 21h
```

Conclusion:

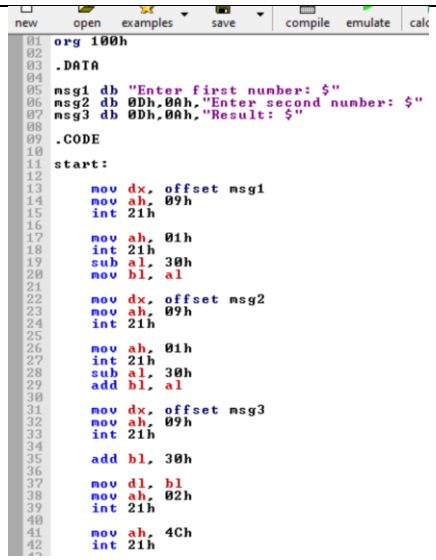
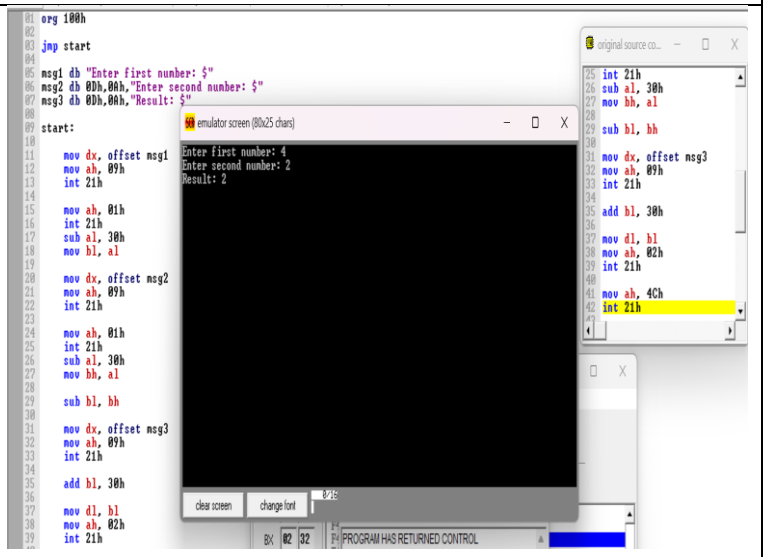
From this experiment, we learned how to perform the addition of two numbers using Assembly Language programming. We also understood the use of registers and arithmetic instructions for processing data. The program was successfully executed in EMU8086 and produced the correct output.

Experiment No: 03

Experiment Name: Write an Assembly Language program to perform subtraction of two numbers.

Introduction:

It is useful for understanding the internal operations of a computer system. In this experiment, an Assembly Language program was written to perform the subtraction of two numbers. The program takes two numbers as input from the user, subtracts one number from another using arithmetic instructions, and displays the result on the screen. This experiment helps to understand arithmetic operations, register manipulation, and input-output techniques in Assembly Language programming.

Code	Output
	

```
org 100h

.DATA

msg1 db "Enter first number: $"
msg2 db 0Dh,0Ah,"Enter second number: $"
msg3 db 0Dh,0Ah,"Result: $"

.CODE

start:

    mov dx, offset msg1
    mov ah, 09h
    int 21h

    mov ah, 01h
    int 21h
    sub al, 30h
    mov bl, al

    mov dx, offset msg2
    mov ah, 09h
    int 21h

    mov ah, 01h
    int 21h
    sub al, 30h
    add bl, al

    mov dx, offset msg3
    mov ah, 09h
    int 21h

    add bl, 30h

    mov dl, bl
    mov ah, 02h
    int 21h

    mov ah, 4Ch
    int 21h
```

Conclusion:

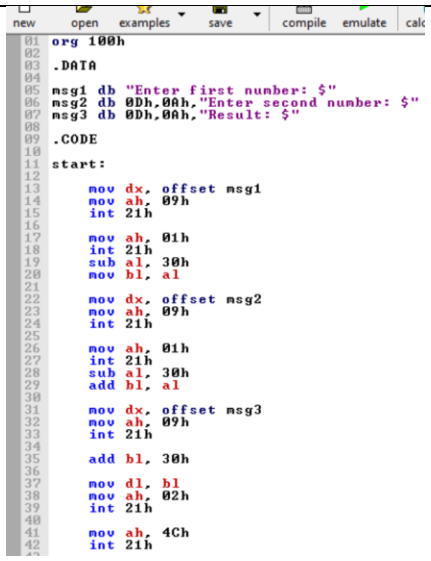
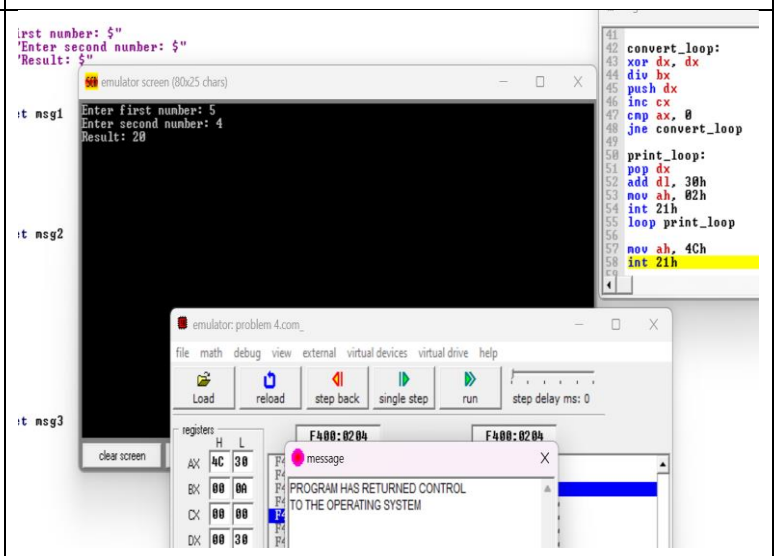
From this experiment, we learned how to perform subtraction between two numbers using Assembly Language. Also gained knowledge about the use of registers, subtraction instructions, and DOS interrupt functions for input and output operations. The program was successfully executed in EMU8086 and produced the expected result.

Experiment No: 04

Experiment Name: Write an Assembly Language program to perform multiplication of two numbers.

Introduction:

It helps programmers understand how arithmetic and logical operations are performed inside the computer. In this experiment, an Assembly Language program was developed to perform the multiplication of two numbers. The program takes two numbers as input from the user, multiplies those using Assembly instructions, and displays the result on the screen. This experiment is important for understanding arithmetic operations, register usage, and data processing in Assembly Language programming.

Code	Output
	

```
org 100h
.DATA

msg1 db "Enter first number: $"
msg2 db 0Dh,0Ah,"Enter second number: $"
msg3 db 0Dh,0Ah,"Result: $"

.CODE

start:

    mov dx, offset msg1
    mov ah, 09h
    int 21h

    mov ah, 01h
    int 21h
    sub al, 30h
    mov bl, al

    mov dx, offset msg2
    mov ah, 09h
    int 21h

    mov ah, 01h
    int 21h
    sub al, 30h
    add bl, al

    mov dx, offset msg3
    mov ah, 09h
    int 21h

    add bl, 30h

    mov dl, bl
    mov ah, 02h
    int 21h

    mov ah, 4Ch
    int 21h
```

```
convert_loop:
xor dx, dx
div bx
push dx
inc cx
cmp ax, 0
jne convert_loop

print_loop:
pop dx
add dl, 30h
mov ah, 02h
int 21h
loop print_loop

mov ah, 4Ch
int 21h
```

Conclusion:

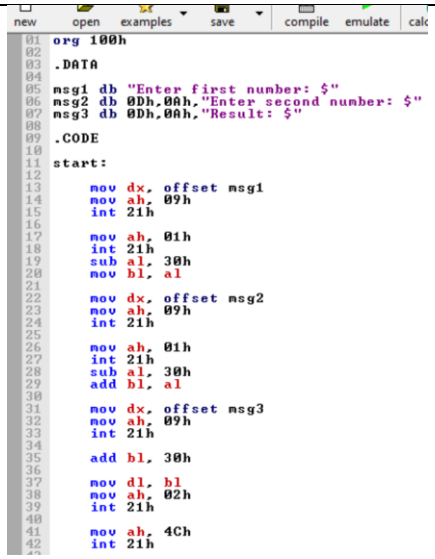
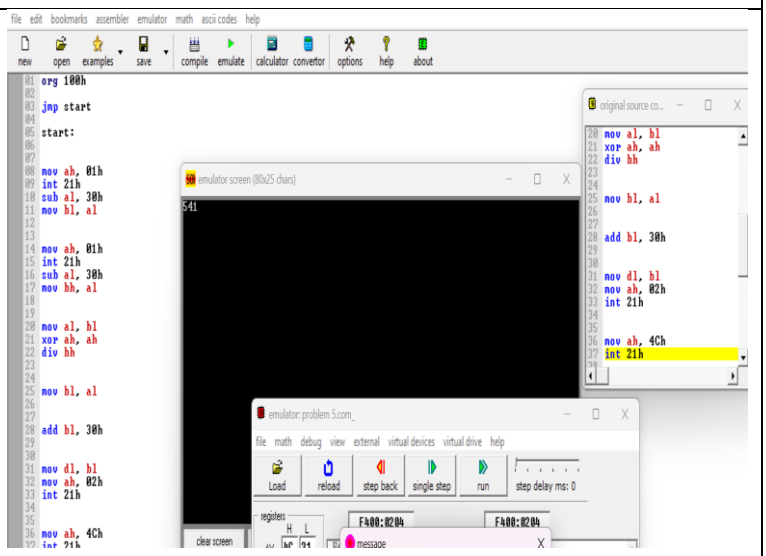
From this experiment, we learned how to perform multiplication of two numbers using Assembly Language. We also understood the use of multiplication instructions, registers, and DOS interrupt functions for input and output operations. The program was successfully executed in EMU8086.

Experiment No: 05

Experiment Name: Write an Assembly Language program to perform division of two numbers.

Introduction:

In this experiment, a program was written to perform the division of two numbers and display the result. This experiment helps to understand division instructions and basic input-output operations in Assembly Language.

Code	Output
	

```
org 100h

.DATA

msg1 db "Enter first number: $"
msg2 db 0Dh,0Ah,"Enter second number: $"
msg3 db 0Dh,0Ah,"Result: $"

.CODE

start:

    mov dx, offset msg1
    mov ah, 09h
    int 21h

    mov ah, 01h
    int 21h
    sub al, 30h
    mov bl, al

    mov dx, offset msg2
    mov ah, 09h
    int 21h

    mov ah, 01h
    int 21h
    sub al, 30h
    add bl, al

    mov dx, offset msg3
    mov ah, 09h
    int 21h

    add bl, 30h

    mov dl, bl
    mov ah, 02h
    int 21h

    mov ah, 4Ch
    int 21h
```

Conclusion:

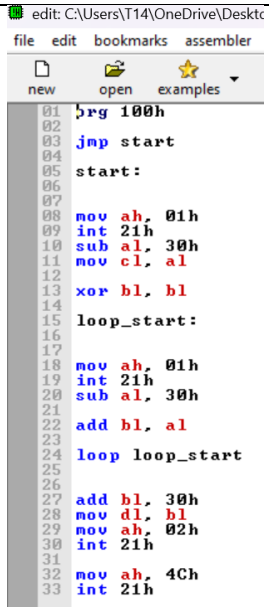
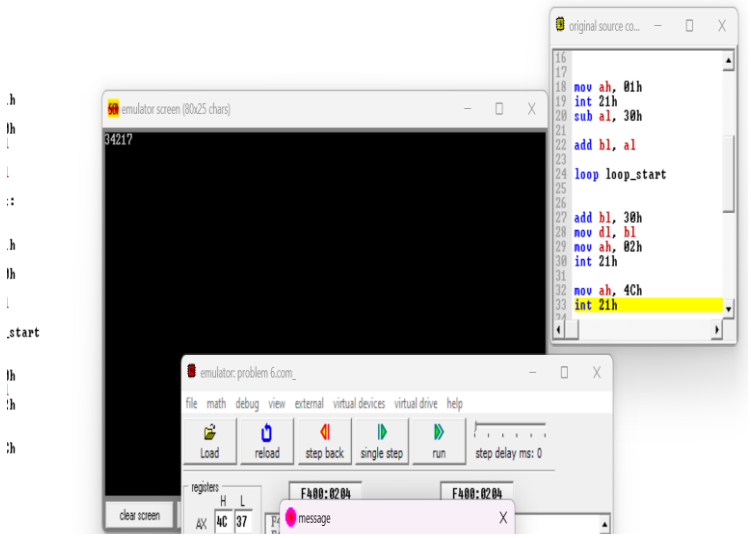
From this experiment, we learned how to divide two numbers using Assembly Language and display the output correctly. The program was successfully executed in EMU8086 and helped us understand basic arithmetic operations and register usage.

Experiment No: 06

Experiment Name: Write an Assembly Language program to take input from the user and display the output.

Introduction:

In this experiment, a program was developed to calculate the sum of N numbers using a loop. The program repeatedly takes input values, adds them using a loop, and displays the final result. This experiment helps to understand loop control, register usage, and arithmetic operations in Assembly Language.

Code	Output
	

```
edit: C:\Users\T14\OneDrive\Deskto
file  edit  bookmarks  assembler
new  open  examples
01  org 100h
02
03  jmp start
04
05  start:
06
07
08  mov ah, 01h
09  int 21h
10  sub al, 30h
11  mov cl, al
12
13  xor bl, bl
14
15  loop_start:
16
17
18  mov ah, 01h
19  int 21h
20  sub al, 30h
21
22  add bl, al
23
24  loop loop_start
25
26
27  add bl, 30h
28  mov dl, bl
29  mov ah, 02h
30  int 21h
31
32  mov ah, 4Ch
33  int 21h
```

Conclusion:

From this experiment, we learned how to calculate the sum of N numbers using a loop in Assembly Language. We also understood how loops and registers are used to perform repeated addition. The program was successfully executed in EMU8086 and produced the correct result, improving our understanding of iterative operations in Assembly Language.

Experiment No: 07

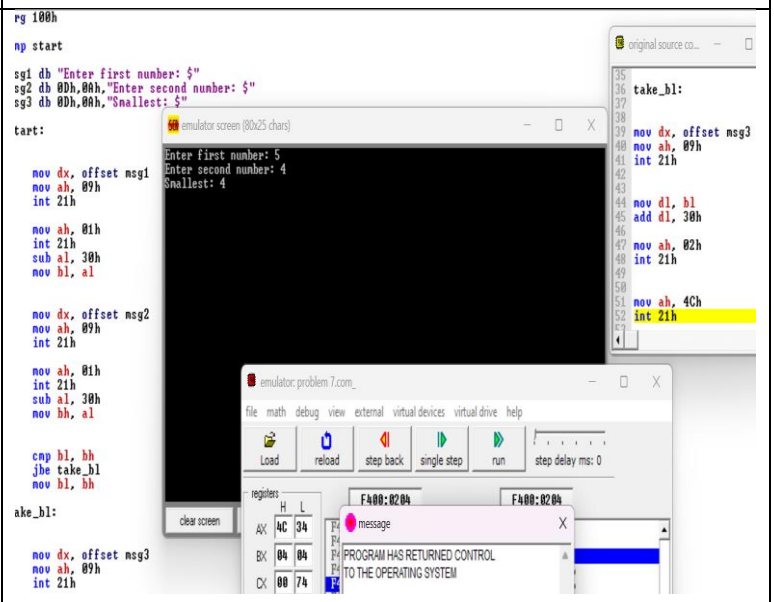
Experiment Name: Write an Assembly Language program to find the smallest number between two given numbers.

Introduction:

In this experiment, a program was written to find the smallest number between two given numbers. The program compares two input values and displays the smaller one as the result. This experiment helps to understand comparison operations and basic decision-making in Assembly Language.

Code	Output

```
org 100h

jmp start

msg1 db "Enter first number: $"
msg2 db 0Dh,0Ah,"Enter second number: $"
msg3 db 0Dh,0Ah,"Smallest: $"

start:

    mov dx, offset msg1
    mov ah, 09h
    int 21h

    mov ah, 01h
    int 21h
    sub al, 30h
    mov bl, al

    mov dx, offset msg2
    mov ah, 09h
    int 21h

    mov ah, 01h
    int 21h
    sub al, 30h
    mov bh, al

    cmp bl, bh
    jbe take_bl
    mov bl, bh

take_bl:

    mov dx, offset msg3
    mov ah, 09h
    int 21h

    mov dl, bl
    add dl, 30h
```



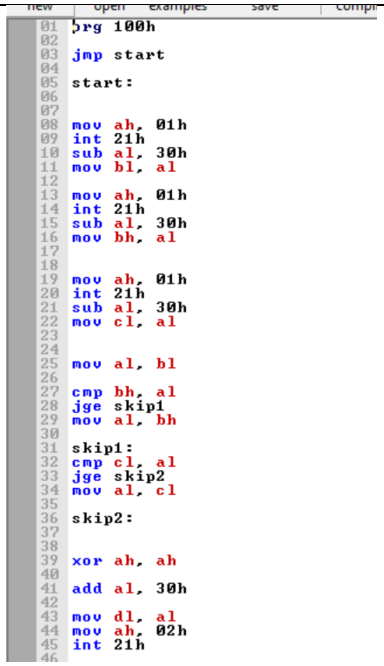
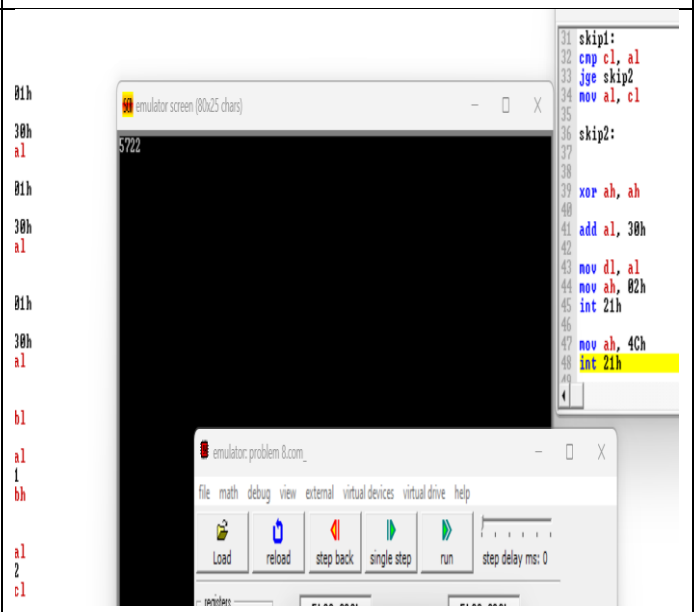
Conclusion:

From this experiment, we learned how to find the smallest number using Assembly Language. We also understood how comparison instructions work to make decisions between values. The program was successfully executed in EMU8086 and produced the correct output.

Experiment No: 08

Experiment Name: Write an Assembly Language program to find the smallest number among three given numbers.

Introduction: In this experiment, a program was written to find the smallest number among three given numbers. The program compares the input values and identifies the smallest one. This experiment helps to understand comparison operations and decision-making in Assembly Language.

Code	Output
	

```
01 org 100h
02
03 jmp start
04
05 start:
06
07
08 mov ah, 01h
09 int 21h
10 sub al, 30h
11 mov bl, al
12
13 mov ah, 01h
14 int 21h
15 sub al, 30h
16 mov bh, al
17
18
19 mov ah, 01h
20 int 21h
21 sub al, 30h
22 mov cl, al
23
24
25 mov al, bl
26
27 cmp bh, al
28 jge skip1
29 mov al, bh
30
31 skip1:
32 cmp cl, al
33 jge skip2
34 mov al, cl
35
36 skip2:
37
38
39 xor ah, ah
40
41 add al, 30h
42
43 mov dl, al
44 mov ah, 02h
45 int 21h
46
```

Conclusion:

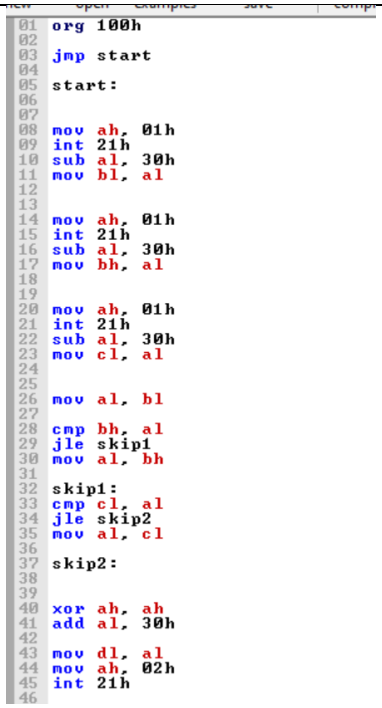
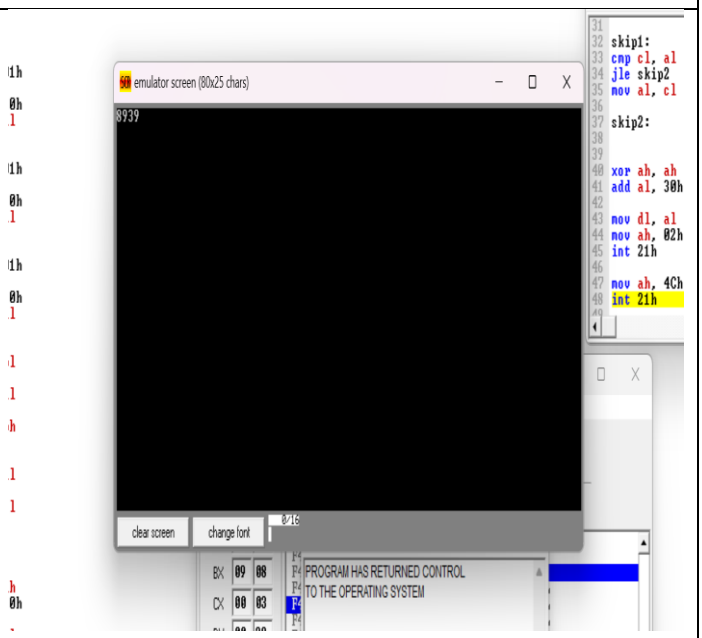
From this experiment, we learned how to find the smallest number among three values using Assembly Language. We also understood how multiple comparisons are performed using conditional logic. The program was successfully executed in EMU8086 and produced the correct result.

Experiment No: 09

Experiment Name: Write an Assembly Language program to find the largest number among three given numbers.

Introduction:

In this experiment, a program was written to find the largest number among three given numbers. The program compares the values and selects the greatest one. This helps to understand comparison and decision-making in Assembly Language.

Code	Output
	

```
org 100h

jmp start

start:


mov ah, 01h
int 21h
sub al, 30h
mov bl, al


mov ah, 01h
int 21h
sub al, 30h
mov bh, al


mov ah, 01h
int 21h
sub al, 30h
mov cl, al


mov al, bl

cmp bh, al
jle skip1
mov al, bh

skip1:
cmp cl, al
jle skip2
mov al, cl

skip2:


xor ah, ah
add al, 30h

mov dl, al
mov ah, 02h
int 21h

```

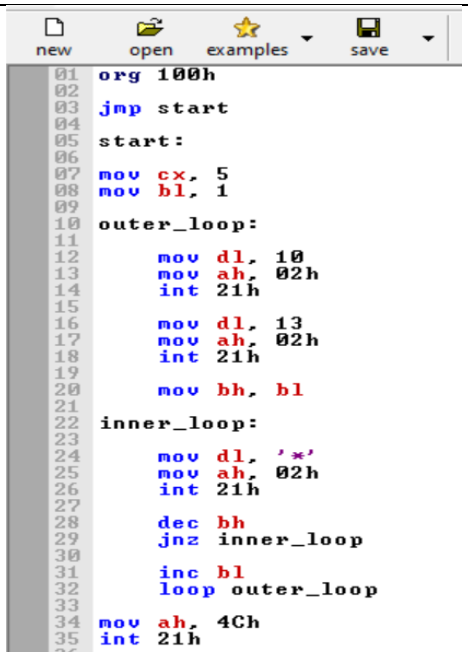
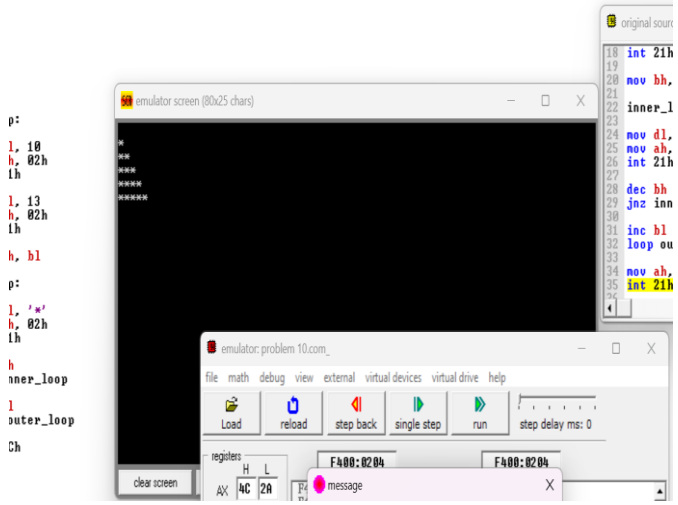

Conclusion: From this experiment, we learned how to find the largest number among three values using Assembly Language. The program was successfully executed in EMU8086 and helped us understand basic comparison operations.

Experiment No: 10

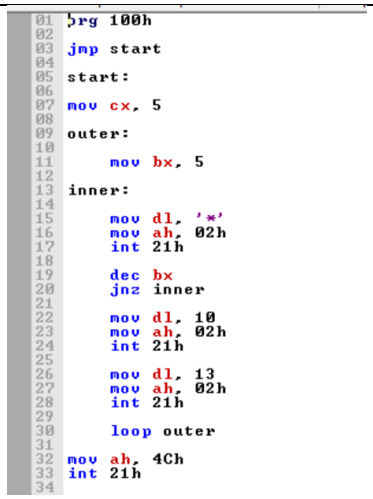
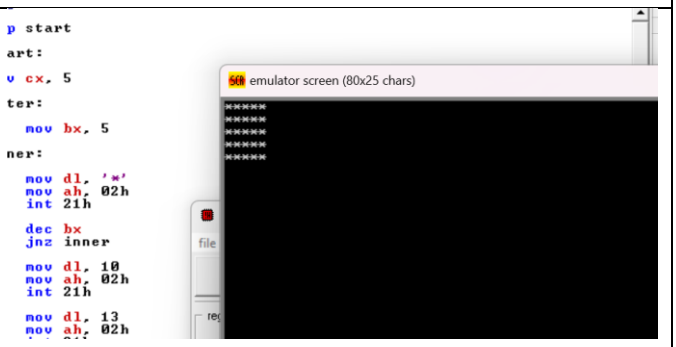
Experiment Name: Write an Assembly Language program to print a simple pattern (e.g., triangle, square, or pyramid).

Introduction:

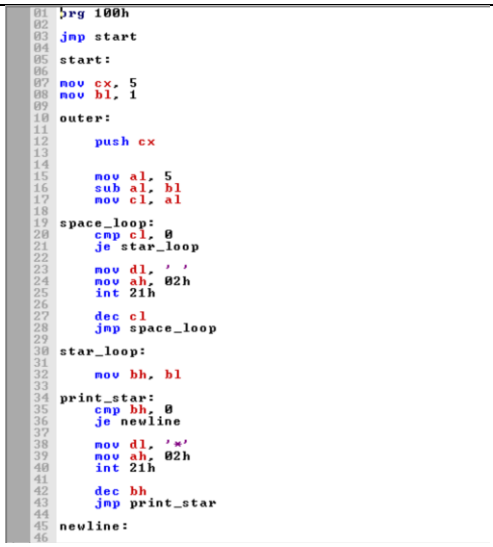
In this experiment, a program was written to print simple patterns like triangle, square, or pyramid. This helps to understand loop usage and output handling in Assembly Language.

Code	Output
	

```
org 100h

jmp start

start:

mov cx, 5
mov bl, 1

outer_loop:

    mov dl, 10
    mov ah, 02h
    int 21h

    mov dl, 13
    mov ah, 02h
    int 21h

    mov bh, bl

inner_loop:

    mov dl, '*'
    mov ah, 02h
    int 21h

    dec bh
    jnz inner_loop

    inc bl
    loop outer_loop

mov ah, 4Ch
int 21h
```

Code(Square)	Output
	

```
org 100h

jmp start

start:

mov cx, 5

outer:

    mov bx, 5

inner:

    mov dl, '*'
    mov ah, 02h
    int 21h

    dec bx
    jnz inner

    mov dl, 10
    mov ah, 02h
    int 21h

    mov dl, 13
    mov ah, 02h
    int 21h

    loop outer

mov ah, 4Ch
int 21h
```


Code(pyramid)	Output
	

```
org 100h

jmp start

start:

mov cx, 5
mov bl, 1

outer:

    push cx


    mov al, 5
    sub al, bl
    mov cl, al

space_loop:
    cmp cl, 0
    je star_loop

    mov dl, ' '
    mov ah, 02h
    int 21h

    dec cl
    jmp space_loop

star_loop:

    mov bh, bl

print_star:
    cmp bh, 0
    je newline

    mov dl, '*'
    mov ah, 02h
    int 21h

    dec bh
    jmp print_star

newline:
```


Conclusion:

From this experiment, we learned how to print simple patterns using loops in Assembly Language. The program was successfully executed in EMU8086 and improved our understanding of iterative programming.